Susanne Klein-Vogelbach

Functional Kinetics

Observing, Analyzing, and Teaching Human Movement

With 329 Illustrations

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Preface to the English Edition

The translation of my book *Funktionelle Bewegungslehre* into English proved an experience unexpectedly rich in learning. When one has no more than a poor command of the language into which one's own work is being translated, one tends - perhaps out of laziness - to assume that one is not competent to judge the result. That was what I thought, anyway.

The English Copy Editing department at Springer-Verlag, in the person of Kersti Wagstaff, taught me better. At her encouragement, armed with dictionaries, I sat down and went through the translation and my German original line by line, making corrections in my stumbling English where the translation did not quite express my intentions. Kersti Wagstaff then spared no efforts in grappling with my ideas so as to be able to rewrite my corrections in clear English. At many places in the manuscript I was told: 'I don't understand what you mean here' - and behold, when I looked again at the German I had to admit that that particular passage could be interpreted in several different ways.

Through this - to me, ideal - process of collaboration, I received invaluable objective criticism of my book, and for this I owe Bernhard Lewerich, who looks after my books at Springer-Verlag, and, especially, Kersti Wagstaff a great debt of thanks.

Bottmingen, August 1989 Susanne Klein-Vogelbach

Preface to the Fourth German Edition

This fourth edition has arisen out of work done on the translation of *Functional Kinetics* into English. Working on that book with Kersti Wagstaff of Springer-Verlag's copy editing department, I was able to go back and improve and clarify many places in the German text.

In addition, the *eight observation criteria of normal gait,* as presented in *Gangschulung zur funktionellen Bewegungslehre* (shortly to be published by Springer-Verlag in the series *Rehabilitation und Prävention,* Vol. 16), have been taken into account:

1. Forward transportation of body segments thorax and head with their frontotransverse axes horizontal and at right angles to the direction of locomotion
2. Step rate
3. Track width
4. Step length
5. Maintenance of the virtual long axis of the body and its vertical alignment
6. Maintenance of the virtual axes of the foot and the alignment of the functional long axis of the foot in the direction of locomotion
7. The walking patterns of body segments legs and pelvis
8. The armswing as *reactio* to the walking patterns of body segments legs and pelvis

My special thanks are due to Bernhard Lewerich of Springer-Verlag, who has met my many requests as an author with great generosity.

Bottmingen, July 1989 Susanne Klein-Vogelbach

Preface to the Third German Edition

The concept of 'functional kinetics' has attracted many supporters and collaborators since its first appearance 7 years ago.
During the teaching of the theory of functional kinetics in training schools and on courses, problems of interpretation arose which were unknown to me, because the concept was my own, and when teaching it myself, I was always able to provide the answer to any queries that came up. I have accepted the queries and problems of my colleagues and students in the spirit of constructive criticism and have appreciated their valuable contribution.
This has led to a complete revision of this book. I have tried to reshape the material in such a way that it is no longer merely an adjunct to my own teaching but an independent presentation of the concept of the observation and analysis of movement on which to base a therapy of movement truly proper to the human being.
This book is intended as an introduction to the theory of functional kinetics. For this reason extensive citation of references has been omitted.

My thanks are due to:
- Springer-Verlag, who made this new edition possible.
- Ortrud Bronner, principal of the School of Physiotherapy at the Kantonspital Basel, who has given a great deal of her time to assist with revision of the manuscript, and Barbara Bartmes, principal of the School of Physiotherapy in Mannheim.
- Katrin Eicke-Wieser for her help in proof-reading and correcting the manuscript, a time-consuming task.
- Adrian and Gudrun Cornford for designing the illustrations for this edition, Verena Sofka for those of the first and second editions, and Ursula Künzle and Ruth Thierstein for the photographs; Andreas Bertram and Jacques Hochstrasser for the presentation of the photographs.
- My colleague Isabella Gloor-Moriconi and all others, too numerous to mention, who have assisted with the preparation of this book.

Basel, November 1983 Susanne Klein-Vogelbach

Preface to the First German Edition

For as long as I can remember, the beauty of movement, human or animal, has fascinated me.

As a teacher of gymnastics I learned to see that healthy and natural movement of a living creature is beautiful, and beautiful movement became my guiding principle in physical education. Confrontation with the reality of the differences in people's gifts for movement, however, has taught me to ask myself why any given sequence of movements should be natural and effortless to some people while others cannot achieve it even after dedicated practice.

Searching for reasons to explain these differences, I was forced to start observing, and began to realize that the constitution, mentality and physical condition of a human being determine whether he is predisposed or indisposed to perform any particular physical activity; suitability (or unsuitability) for any type of movement must, therefore, be predictable. Thus I came to appreciate how relative the concepts 'normal', 'healthy' and 'sick' are. The step to therapy was small and changed my concept of movement training only gradually. My model remained the same; only the road to achieving it became longer, sometimes more arduous, and on occasion impassable.

In this book I have tried to order the experience of a lifetime as a professional teacher of movement and to point towards a possible way of considering movement systematically, of observing and analysing it, and of conveying the findings in such a manner that the patient can learn from them.

We should never forget that human understanding is insufficient to grasp the phenomena of movement in their entirety. But we can observe them and come to recognize their outward form.

My thanks for advice and collaboration to: Georg Klein-Vogelbach, my husband; Gisela Rolf and Irmgard Flückiger; Verena Sofka-Lagutt, graphics; Ortrud Bronner, Katrin Eicke-Wieser, Verena M. Jung, readers; and Heidi Säckinger-Wolf and Anne Schäfer, secretaries.

Basel, February 1976 Susanne Klein-Vogelbach

Note on Pronouns

For clarity and brevity, 'she' has been assigned to the physiotherapist and 'he' to the patient.

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